Host identity is a dominant driver of mycorrhizal fungal community composition during ecosystem development

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Summary

- Little is known about the response of arbuscular mycorrhizal fungal communities to ecosystem development. We use a long-term soil chronosequence that includes ecosystem progression and retrogression to quantify the importance of host plant identity as a factor driving fungal community composition during ecosystem development.
- We identified arbuscular mycorrhizal fungi and plant species from 50 individual roots from each of 10 sites spanning 5–120 000 yr of ecosystem age using terminal restriction fragment length polymorphism (T-RFLP), Sanger sequencing and pyrosequencing.
- Arbuscular mycorrhizal fungal communities were highly structured by ecosystem age. There was strong niche differentiation, with different groups of operational taxonomic units (OTUs) being characteristic of early succession, ecosystem progression and ecosystem retrogression. Fungal alpha diversity decreased with ecosystem age, whereas beta diversity was high at early stages and lower in subsequent stages. A total of 39% of the variance in fungal communities was explained by host plant and site age, 29% of which was attributed to host and the interaction between host and site (24% and 5%, respectively).
- The strong response of arbuscular mycorrhizal fungi to ecosystem development appears to be largely driven by plant host identity, supporting the concept that plant and fungal communities are tightly coupled rather than independently responding to habitat.

Introduction

Understanding the process of ecosystem development and the mechanisms driving species coexistence is a major topic in ecology (Clements, 1936; Walker & Chapin, 1987; Richardson *et al.*, 2005; Peltzer *et al.*, 2010). Recent work has highlighted the important role of plant–soil interactions driving diversity patterns during plant succession (Fukami & Nakajima, 2013; Jangid *et al.*, 2013), yet the role of plant–mycorrhizal fungal interactions during succession and ecosystem development is less clear (Dickie *et al.*, 2013). Long-term soil chronosequences (i.e. sites differing in soil age, but with similar parent material, climate and regional species pools; Jenny, 1980) are an ideal tool for resolving how plant–fungal symbioses respond to pedogenesis (Wardle *et al.*, 2008). Strong gradients of soil nitrogen (N) and phosphorus (P) during ecosystem development result in consistent shifts from N to P limitation for plant species and ecosystem processes (Richardson *et al.*, 2004; Allison *et al.*, 2007). Retrogression then occurs in the later stages when the impoverishment of soil nutrients leads to a decline in ecosystem process rates (Peltzer *et al.*, 2010).

Mycorrhizal fungi have an active role in nutrient cycling, acquiring mineral P, facilitating the decomposition of organic nutrients (Hodge *et al.*, 2001) and transferring nutrients to plant partners. In light of the strong connectivity between mycorrhizal fungi, plants and nutrient cycling, we expect mycorrhizal fungi to both respond to and potentially drive ecosystem development along chronosequences (Dickie *et al.*, 2013). Although there may be shifts between mycorrhizal types in some successions (i.e. from non-mycorrhizal to arbuscular mycorrhizal to ectomycorrhizal; Read, 1993; Lambers *et al.*, 2008), arbuscular mycorrhizal plants are dominant throughout early, mature and retrogressive stages in most major soil chronosequences (Dickie *et al.*, 2013). This suggests that changes in fungal species composition within mycorrhizal types may be critical in the understanding of mycorrhizal functioning during ecosystem development and retrogression (Dickie *et al.*, 2013; Sikes *et al.*, 2014).

Several studies have documented changes in arbuscular mycorrhizal fungal communities during early succession and mature ecosystems (Dickie *et al.*, 2013). Newly exposed substrates support unpredictable fungal communities as a result of the stochastic immigration of propagules from surrounding landscapes

(Jumpponen *et al.*, 2012; Dickie *et al.*, 2013). During ecosystem progression, ecological sorting by biotic and abiotic environmental filters may select successful species (Jumpponen *et al.*, 2012), leading to species niche differentiation. Despite this, few clear patterns of fungal species turnover have been detected during ecosystem progression across studies, and no previous study has looked at changes in arbuscular mycorrhizal fungal communities throughout longer term ecosystem development spanning from early primary succession to retrogressive stages (Dickie *et al.*, 2013). Based on the strong shifts during ecosystem development in the availability and forms of soil N and P (Richardson *et al.*, 2004; Allison *et al.*, 2007) and changes in plant hosts and plant root traits (Richardson *et al.*, 2004; Holdaway *et al.*, 2011), we hypothesized that arbuscular mycorrhizal fungal species would show niche differentiation for different ecosystem ages (Hypothesis 1). In particular, we expected that deterministic processes would lead to the environmental filtering of fungal communities, resulting in individual species increasing, decreasing or showing unimodal responses to ecosystem age.

Plant communities increase in species richness during ecosystem progression and decrease during ecosystem retrogression (Richardson *et al.*, 2004; Wardle *et al.*, 2008), but patterns of arbuscular mycorrhizal fungal diversity are not well known (Dickie *et al.*, 2013). A prediction would be that high-alpha-diversity plant communities would support high-alpha-diversity fungal communities (Helgason *et al.*, 2002; Dickie, 2007). However, arbuscular mycorrhizal fungal propagules are likely to be dispersal limited (Dickie *et al.*, 2013), which might contribute to early successional communities being compositionally determined by the stochastic process of mycorrhizal fungal propagule deposition in addition to the deterministic effects of plant community composition. As such, we would expect a higher variability, or beta diversity (Anderson *et al.*, 2011), of fungal communities within early successional sites relative to older sites. During primary succession, the accumulation of propagules over time may increase alpha diversity (Wu *et al.*, 2007), whereas environmental filtering may cause a decrease in beta diversity (Margalef, 1963). Finally, particularly strong environmental filtering during retrogression may reduce both alpha and beta diversity. On this basis, we hypothesized that arbuscular mycorrhizal fungal species richness (alpha diversity) would respond to pedogenesis in parallel with plant communities (increasing and then decreasing), and that beta diversity (variability and turnover; Anderson *et al.*, 2011) would be highest at the earliest ecosystem stages and decline during ecosystem progression and retrogression (Hypothesis 2).

Although we expected fungal communities to change during ecosystem development, an important remaining question in mycorrhizal ecology is the mechanism driving these changes. Zobel & Öpik (2014) suggested that the mechanisms determining plant and fungal communities may change over successional time. They argued that successional communities change according to the Passenger or Driver hypotheses, with plants driving fungal communities or vice versa (Hart *et al.*, 2001), whereas, in successionally stable, climax-like vegetation types, both plants and arbuscular mycorrhizal fungi respond independently to habitat conditions, and neither partner drives changes in the other (Habitat hypothesis; Zobel & Öpik, 2014). In a study of primary succession in sand dunes, Sikes *et al.* (2014) found that only soil characteristics, and not host plant identity, determined mycorrhizal fungal growth patterns (arbuscules vs hyphae). This may provide some support to the Habitat hypothesis (despite not directly measuring fungal community composition). Conversely, a comparison of arbuscular mycorrhizal fungi in younger and older forest ecosystems found increased host specialization in the older forest, whereas plants in younger forests were associated with generalist fungi (Bennett *et al.*, 2013). Zobel & Öpik (2014) did not discuss retrogressive ecosystems, and no previous studies have examined arbuscular mycorrhizal fungi in retrogressive successions. We suggest that a decline in available soil nutrient resources during retrogression may increase the relative differences in efficiency among fungal partners from a plant perspective (Kiers *et al.*, 2011). This may therefore increase the importance of Passenger/Driver mechanisms in retrogressive ecosystems, consistent with increased specialization of arbuscular mycorrhizal fungi in the old-growth forests observed by Bennett *et al.* (2013). On this basis, we hypothesized that host plant identity would be a strong mechanism driving mycorrhizal fungal community composition with ecosystem age, but least important in mature, relatively high-nutrient-availability ecosystems (Hypothesis 3).

The testing of Hypothesis 3 requires the unravelling of the relative importance of host identity vs soil habitat in structuring fungal communities, which is a difficult task in natural ecosystems as plant hosts change along with habitat (Vandenkoornhuyse *et al.*, 2003; Zobel & Öpik, 2014). In this study, we used a well-characterized long-term soil chronosequence (Franz Josef glacier, New Zealand) to test for shifts in arbuscular mycorrhizal fungal community structure throughout 120 000 yr of ecosystem development. We measured local (alpha) fungal diversity within sites differing in ecosystem age, and community divergence and species turnover (beta diversity, Anderson *et al.*, 2011) among sites to characterize fungal community dynamics with ecosystem development (testing Hypotheses 1 and 2, above). Although plant communities change across the chronosequence, many plant species occur at more than one site (Richardson *et al.*, 2004). By determining the arbuscular mycorrhizal fungal community present in nearly 445 individual plant root fragments, 86% of which represented plant species found at more than one site age, we were able to directly test the importance of host plant identity after accounting for the effects of soil habitat (site age) as two critical factors structuring fungal communities (testing Hypothesis 3).

Materials and Methods

Chronosequence description

The Franz Josef chronosequence in southern New Zealand is a series of schist outwash surfaces that date from the present to *c.* 120 000 yr since glacial retreat. High rainfall (up to 8 m yr^{-1}), an oceanic climate and perhumid conditions supporting year-round soil microbial activity result in rapid nutrient loss and pedogenesis (Almond *et al.*, 2001). The youngest stages (< 15 yr) are characterized by open turf vegetation and shrubland. Evergreen rainforest species colonize sites within 65 yr, reaching tall forest communities (> 30 m) and peak biomass at the 5000-yr-old site. High biomass and tall forests are maintained until the 12 000-yr-old site (Wardle *et al.*, 2004). Later stages (60 000–120 000 yr) exhibit signs of retrogression as species richness, forest height and biomass decline to an open heath forest (Richardson *et al.*, 2004; Wardle *et al.*, 2004). Angiosperms dominate early in succession, and conifers (Podocarpaceae) progressively co-dominate and ultimately dominate as soil nutrients are depleted (Richardson *et al.*, 2004; Holdaway *et al.*, 2011). Arbuscular mycorrhizal associations dominate through the chronosequence with no presence of ectomycorrhizas (Dickie *et al.*, 2013). Total P sharply declines from 900 mg kg^{-1} at the 15-yr-old site to 100 mg kg^{-1} by the oldest stage, whilst also shifting from inorganic to increasingly recalcitrant and organic forms, whereas total mineral soil N peaks at 9 g kg^{-1} within 500 yr of ecosystem development and then declines to *c.* 3 g kg^{-1} (Table 1; Richardson *et al.*, 2004). Full soil chemistry is described in Richardson *et al.* (2004).

We selected 10 sites across the full range of ecosystem ages, including seven from Richardson *et al.* (2004), with a new site at 5 yr which replaces the 5-yr site of Richardson (now our 15-yr site) and new sites at 1000 and 12 000 yr (Table 1). Mineral and organic soil samples were collected at early-stage sites (5 and 15 yr) and at the newly established 1000- and 12 000-yr sites following the same approach as Richardson *et al.* (2004) (Table 1 and Supporting Information Table S1). Soil chemistry at other sites was not re-characterized for the following reasons: we assume that no significant changes related to soil weathering have occurred in the last decade in these older sites; we used the same accredited laboratory which employs standardized calibration soils to maintain consistency over time; and the data are only provided as site description and not used in any statistical analysis.

Root and leaf sampling

The arbuscular mycorrhizal fungal community was characterized from 50 root samples collected at each one of the 10 sites. Each sample comprised a single root fragment, 10–20 cm in length and < 2 mm in diameter. At each site, we established two parallel 50-m transects separated by 4 m, with a randomly determined starting point. We took two root samples every 2 m along each transect. To ensure that the full community of roots and fungi was sampled, we collected equal numbers of roots from each of three predetermined, randomly selected depth classes (middle of the organic horizon or top 0–10 cm of soil if no organic present; top of mineral horizon; and 10–20 cm of mineral horizon). Roots were washed to remove attached soil, stored at 4°C for up to 10 d and freeze-dried. The second sample from each point was only utilized if the first sample failed in molecular analysis. Leaf samples of most plant species present at each chronosequence site were identified, collected and freeze-dried as reference samples for DNA identification of roots.

Arbuscular mycorrhizal fungal amplification and sequencing

Two complementary molecular analysis techniques were used to characterize the arbuscular mycorrhizal fungal community colonizing the roots: pyrosequencing on samples pooled within the site ($n = 10$) and terminal restriction fragment length polymorphism (T-RFLP) on each individual root sample ($n = 500$). Pyrosequencing was used to describe the arbuscular mycorrhizal fungal community per site, whereas T-RFLP was used to describe the community at sites and within roots per site. For both techniques, Glomeromycota DNA was amplified using the universal eukaryotic primer NS31 (Simon *et al.*, 1992) and the arbuscular mycorrhizal fungal-specific primer AML2 (Lee *et al.*, 2008). A subsample of 0.15 mg of each root was ground in a bead beater and then extracted with a PowerSoil DNA Isolation Kit (Mo Bio Labs, Carlsbad, CA, USA).

Pyrosequencing

The arbuscular mycorrhizal fungal-specific primers linked to sequencing adapters (Roche Diagnostics NZ Ltd, Auckland, New Zealand) and to multiplex identifier (MID) barcode sequences were used to amplify 1 μl of DNA extract with 0.4 μM

Table 1 Site characteristics and ages along the Franz Josef chronosequence (mean ± SE) from the glacial forefront (< 5 yr) through peak biomass (5000–12 000 yr) and retrogression (modified from Richardson *et al.*, 2004; soil chemical analysis was performed at the Environmental Chemistry Laboratory, Landcare Research, Palmerston North, New Zealand)

Site age (yr)	Precipitation (mm)	Elevation (m)	Total phosphorus (mg kg^{-1})	Total nitrogen (%)	Total carbon (%)	pH
< 5	6520	240	774.9 ± 21.7	0.01 ± 0.00	0.05 ± 0	8.14 ± 0.06
15	6576	240	910.6 ± 27.8	0.04 ± 0.01	0.62 ± 0.17	6.44 ± 0.13
70[a]	6188	220	804.8 ± 36.3	0.22 ± 0.04	3.29 ± 0.53	5.62 ± 0.1
290[a]	6300	200	513.6 ± 36.3	0.61 ± 0.08	10.69 ± 1.16	4.36 ± 0.13
500[a]	6278	200	457.7 ± 29.1	0.80 ± 0.16	14.6 ± 2.85	3.95 ± 0.1
1000	6390	210	210.4 ± 25.6	0.15 ± 0.02	3.09 ± 0.32	4.63 ± 0.03
5000[a]	6427	245	326.6 ± 20.8	0.50 ± 0.04	9.39 ± 0.75	3.87 ± 0.03
12 000	3623	225	206.1 ± 29.3	0.40 ± 0.06	8.78 ± 1.26	4.15 ± 0.05
60 000[a]	3706	265	200.9 ± 16.2	0.34 ± 0.04	8.62 ± 1.36	3.87 ± 0.06
120 000[a]	3652	145	107.8 ± 10.9	0.36 ± 0.03	9.01 ± 1.37	3.93 ± 0.04

[a]Information from Richardson *et al.* (2004). The 15-yr site is the same as the 5-yr site in Richardson *et al.* (2004), but resampled for soil chemistry.

of each primer, 0.2 mM deoxynucleoside triphosphates (dNTPs), 2 mM $MgCl_2$, 0.2 mg ml^{-1} BSA (New England BioLabs, Auckland, New Zealand) and 0.05 U μl^{-1} Fast Taq DNA polymerase (Roche Diagnostics NZ Ltd). Thermal cycling for the PCRs began with an initial denaturing step at 94°C for 3 min, followed by 35 cycles of 30 s at 94°C, 1 min at 58°C and 1 min at 72°C, and ended with a final extension step of 72°C for 10 min. PCR products from each site were amplified with the same MID tag and evenly pooled. Equimolar amounts of each site PCR mixture were combined for pyrosequencing on a GS Junior (Roche Applied Science) at Landcare Research (Auckland, New Zealand).

T-RFLP and sequencing

Fluorochrome-labelled primers NS31-VIC and AML2-6FAM were used to amplify arbuscular mycorrhizal fungal DNA on each root sample employing the same protocols as for pyrosequencing. Five per cent of the PCRs did not amplify. For those samples, a nested PCR was used, combining a first PCR with NS1 and NS4 (universal fungal primers: White *et al.*, 1990) followed by NS31-VIC and AML2-6FAM. Universal fungal primers were concentrated at 0.1 µM and the annealing temperature was set at 40°C. Amplicons were purified using a DNA Clean & Concentrator Kit (Zymo Research Corporation, Nelson, New Zealand). Root DNA used in T-RFLP and pyrosequencing analysis came from the same root samples.

Virtual digestions of 189 commercial restriction enzymes were run on the 10 most abundant fungal sequences resulting from pyrosequencing, and the enzymes HinfI (Life Technologies NZ Ltd., Auckland, New Zealand) and MslI (New England BioLabs) were selected as having the greatest ability to distinguish operational taxonomic units (OTUs). Clean amplicons were digested for 6 and 1 h, respectively, and denatured using HiDi formamide mixed with MapMarker 1000 ROX standard (BioVentures, Murfreesboro, TN, USA). Terminal restriction fragments (TRFs) were separated by capillary electrophoresis using a Prism 3100 Genetic Analyzer (Applied Biosystems) at the University of Canterbury Sequencing Centre (Christchurch, New Zealand).

Plant amplification and sequencing

The plant identity of each root was determined by sequencing and matching to DNA sequences from leaf samples in GenBank. Plant DNA from leaves was extracted using the Intron Plant DNA extraction kit (Intron Biotechnology, Gyeonggi-Do, South Korea). Plant DNA from roots (same extract as for fungi) and leaves was amplified using the chloroplast *trn*L (UAA) gene primers c and d (Taberlet *et al.*, 1991). These primers failed to amplify DNA from fern species, and hence specific fern primers trnL-f and trn-Fern (Trewick *et al.*, 2002) were used to amplify the DNA in root samples in which PCR failed using the trnL-c-d primers and for fern leaves. Amplicons from leaf and root samples were purified using ExoI and Sap enzymes (Thermo Fisher Scientific, Auckland, New Zealand) and sequenced (University of Canterbury Sequencing Centre, Christchurch, New Zealand). Table S2 shows the number of roots per plant species at each site.

Bioinformatics

Pyrosequencing raw sequences were processed by combining R (R Development Core Team, 2013) and the Quantitative Insights Into Microbial Ecology (QIIME) pipeline (Caporaso *et al.*, 2010). Using R, sequences were assigned to sites according to MID tags, trimmed for primers and MID tags, sequences shorter than 300 bp were removed, and homopolymers > 3 bp in length were collapsed to a length of 3 bp to improve accuracy of pyrosequencing data (Huse *et al.*, 2007). Using QIIME, sequences were quality checked, clustered into OTUs and a representative sequence per OTU was chosen. Different similarity cutoffs were tested (91–99.5%). We chose the 95% similarity cutoff after observing an exponential increase in the number of OTUs at higher similarities. Using a cutoff of 97% would have resulted in one study site having 351 OTUs, which seemed unreasonable compared with a total known global species number of 241 in Schüßler & Walker (2010) and to 341 virtual taxa in the MaarjAM database (Öpik *et al.*, 2013). We also found that the fungal community from our study showed the same ecological patterns regardless of clustering at 95% or 97% (Fig. S1), supporting the suggestion that OTU delineation does not have large effects on observed outcomes (Lekberg *et al.*, 2014; Powell & Sikes, 2014). Uchime was used to detect and remove chimeric sequences (Edgar *et al.*, 2011). Representative pyrosequencing OTUs were submitted to the Sequence Read Archive (SAR: SAMN02998060–SAMN02998069).

The taxonomic identity of pyrosequencing OTUs was determined by BLAST (Camacho *et al.*, 2009) comparison against sequences contained within the MaarjAM database (Öpik *et al.*, 2010, 2014) using BLAST on homopolymer-collapsed sequences. We considered a match at ≥ 95% similarity over 220 bp to represent a species, ≥ 90% a genus and ≥ 80% a family. Singletons (OTUs represented by a single observation) were removed. See Supplementary Information Notes S1 for further details on OTU clustering and recovery method.

The TRAMPR package in R (Fitzjohn & Dickie, 2007) was used to process T-RFLP data. Peaks with a height ≤ 20% of the largest peak were ignored as background noise. The build.knowns function from the TRAMPR package was used to define T-RFLP-OTUs by the highest peak from each restriction enzyme direction (forward and reverse).

Plant species were identified based on BLAST matches to GenBank at 98% similarity, including sequences from our vouchered plant leaf collections (GenBank: KF591217–KF591341).

Statistical analysis

We obtained two complementary datasets, one from T-RFLP and one from the pyrosequencing analysis. Pyrosequencing data, at the level of site, were used to test for niche differentiation, to explore diversity patterns and to identify arbuscular mycorrhizal fungal species. T-RFLP data, at the level of 500 individual roots, were used to quantify the effect of host identity and to assess diversity patterns. We treated site age as a continuous variable across the chronosequence. Site age was also treated as a categorical variable as reported in the Supporting information (no qualitative difference in results was found).

Mycorrhizal fungal communities among sites were compared by multivariate analysis using the *vegan* package in R (Oksanen *et al.*, 2013). Compositional differences among samples were calculated using the Bray–Curtis dissimilarity measure. Nonmetric multidimensional scaling (NMDS) was used to graphically visualize the organization of samples in two-dimensional space employing the functions *metaMDS* and *ordiplot* in the *vegan* package.

We compared linear, log-linear and quadratic models to test whether individual OTUs showed evidence of niche differentiation, that is, significant relationships between site age and OTU abundance. Significant linear or log-linear decreases with site age were taken as evidence of early successional fungal communities, and significant linear or log-linear increases with site age were taken as evidence of fungi specialized on the late stages of retrogression. Significant quadratic relationships (increase followed by decline) were taken as evidence of fungi showing significant correlation with mature (peak biomass) ecosystem stages. Abundance was quantified as the proportion of pyrosequencing sequences representing that OTU within a site. Empirical evidence suggests that pyrosequencing read counts are sufficiently quantitative to detect major changes in relative abundances (Kauserud *et al.*, 2012). The five models were tested for each pyrosequencing OTU (OTU abundance vs site age). From the five tested models, that with the lowest Akaike Information Criterion (AIC) was selected for each OTU, and considered to be significant if $P < 0.05$. The large number of taxa tested separately here could potentially inflate the type I error rate, such that it would be expected that some would show significant responses to chronosequence age as a result of chance alone. Permutational multivariate analysis of variance (PerMANOVA), conducted before individual taxa tests, ensured that there was an overall significant effect before testing individual taxa. Further, we used a Bernoulli process (Moran, 2003) to calculate the probability of finding the observed number of significant results by chance, given the number of tests. Finally, the tests of individual taxa are likely to be a conservative underestimate, as the limited range of models we attempted did not encompass all possible abundance curves.

We analysed the response of mycorrhizal fungal alpha and beta diversity to ecosystem development. Alpha diversity was defined as the number of OTUs at each site after rarefaction to 1000 reads per site for pyrosequencing and to 100 OTU counts per site for T-RFLP. Beta diversity was defined following Anderson *et al.* (2011) as 'variation beta diversity', which is the variability in OTU composition among roots within a site, and 'turnover beta diversity', which is the change in OTU composition between consecutive sites along the chronosequence. Variation beta diversity was calculated using the functions *betadisper* and *permutest.betadisper* in the *vegan* package as the average distance of the T-RFLP samples to the group (site) centroid in a multivariate space. Turnover beta diversity was calculated as the pairwise dissimilarity measure among consecutive sites for both pyrosequencing and T-RFLP data.

The effects of soil habitat (site age) and host identity on the fungal community were tested with PerMANOVA using the function *adonis* in the *vegan* package, which allows the fitting of linear models to partition distance matrices among sources of variation. The effect of site age (log transformed) was tested on the basis of both pyrosequencing and T-RFLP data, and host identity was tested on the basis of T-RFLP data. Sequential sums of squares in PerMANOVA were used to measure the effect size of each term (site age and host identity) after accounting for the variance explained by the other (Schmid *et al.*, 2002). The effect of host identity within each of the 10 sites was individually tested using PerMANOVA. In the same way, the effect of site age was tested on all individual plant hosts in each of two or more sites three or more times (including only those sites).

In addition, we used NMDS to visualize changes in the sample plant community across the chronosequence, and calculated plant species richness and turnover beta diversity (Fig. S1). As plant community data are based on 50 roots, the sample is biased towards plant species with a more extensive root system, and is provided as context for interpretation. Aboveground plant community responses to ecosystem development are available from previous studies (Richardson *et al.*, 2004; Holdaway *et al.*, 2011).

Results

Fungal community composition

Pyrosequencing detected 113 arbuscular mycorrhizal fungal OTUs from the 10 pooled root samples across the chronosequence (Fig. S2). T-RFLP detected 34 OTUs across the 498 individual root samples from which PCR products were obtained, with 3.0 ± 0.08 T-RFLP OTUs (mean and standard error) per root fragment (details of pyrosequencing data, bioinformatics process and distribution by ecosystem ages and Glomeromycota families are given in Notes S1).

Mycorrhizal fungal communities were strongly structured by ecosystem age (Fig. 1). Pyrosequencing data showed a strong effect of site age on arbuscular mycorrhizal fungal community composition (PerMANOVA, $F_{1,9} = 5.51$, $R^2 = 0.41$, $P < 0.01$; Fig. 1a). The NMDS ordination revealed that this effect reflected large changes in composition along one axis from 5 to 15 and 15 to 70 yr (i.e. primary succession to an angiosperm plant community), a major orthogonal shift from 70 to 280 yr (i.e. transition to mature forest) and then relative stability from 280 to 5000 yr, after which there was another large shift to a new, relatively stable set of communities from 12 000 to 120 000 yr (i.e. dominance by gymnosperms and the beginning of retrogression) (Fig. 1a).

Niche differentiation of the fungal community

At the level of individual OTUs, 27 of the 75 pyrosequencing OTUs found in more than two sites showed significant niche differentiation with ecosystem age (significant linear, log-linear or quadratic regressions). Of these OTUs, 12 ('early OTUs')

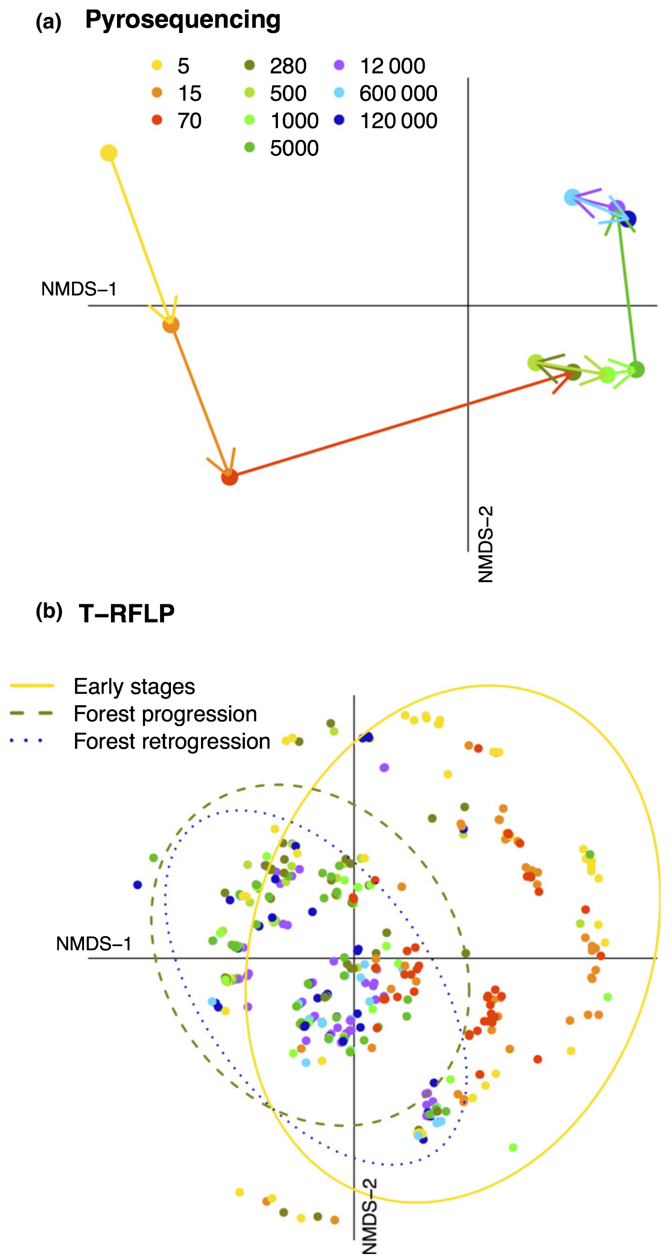

Fig. 1 Nonmetric multidimensional scaling (NMDS) of arbuscular mycorrhizal fungal communities along the 120 000 yr of the Franz Josef chronosequence. (a) Pyrosequencing data (n = 10, final stress = 0.25); each dot represents a chronosequence site. (b) Terminal restriction fragment length polymorphism (T-RFLP) data (n = 498, final stress = 0.18); each dot represents a single sample root. Ellipses delimit the 90% confidence interval around centroids for early successional stages (5–70-yr-old sites), forest ecosystem progression (290–1000-yr-old sites) and during ecosystem retrogression (60 000–120 000-yr-old sites).

showed a significant linear or log-linear decline in abundance with ecosystem age (Fig. 2a). These early OTUs were absent or nearly absent after 70 yr of ecosystem development. A second group of 10 OTUs ('mature OTUs') were best described using quadratic regressions, as they increased in abundance from early stages, reached a peak abundance between 1000 and 5000 yr, and decreased in abundance during retrogression (Fig. 2b). Finally, five OTUs ('retrogressive OTUs') showed a linear or log-linear increase in abundance from early stages to retrogressive stages (Fig. 2c). Fungi in the families Archaeosporaceae and Diversisporaceae only occurred as early OTUs, and fungi in the family Acaulosporaceae only within the mature OTU group. The Glomeraceae was represented in all three OTU groups, and included all of the retrogressive OTUs. We did not test niche preference in the 35 OTUs occurring in two or fewer sites ('rare OTUs'), including 16 OTUs found in a single site only, given the expectation of little statistical power to detect shifts. The probability of finding 28 significant responses from the 75 taxa tested was $P = 1.1 \times 10^{-16}$ (Bernoulli process analysis).

Fungal alpha and beta diversity through ecosystem development

Early successional communities showed higher arbuscular mycorrhizal fungal richness (alpha diversity, Fig. 3a) and higher within-site variation (variation beta diversity, Fig. 3b) than did the later stages of progression or retrogression. Alpha diversity was highest at 15- and 70-yr-old sites, with the two lowest alpha diversity values for pyrosequencing data observed at the two oldest sites, coinciding with retrogressive stages. Variation beta diversity was high at 5- and 15-yr sites (Fig. 3b), but was lower and relatively constant thereafter.

Arbuscular mycorrhizal fungal species turnover between consecutive sites (turnover beta diversity) was highest between the three youngest sites, peaking at the transition between 70 and 290 yr when tall forest canopies developed (Fig. 3c). Pyrosequencing data showed a second peak in turnover beta diversity between 5000 and 12 000 yr; this latter peak was not reflected by the T-RFLP data (Fig. 3c). A similar pattern was apparent when graphically representing pairwise distances among all the sites (Fig. 3d), with large changes in early-stage communities, followed by long periods of relative stability. Again, pyrosequencing and T-RFLP showed similar patterns during early stages, but pyrosequencing distinguished retrogressive from mature ecosystems, whereas T-RFLP did not (Fig. 3c,d).

Quantification of the effect of plant host on fungal communities

PerMANOVA on T-RFLP data from individual root samples explained 39% of the variance in fungal communities based on R^2 values (Table 2), similar to the 41% explained by site age for pyrosequencing data. Using sequential sums of squares, and running the model twice, each time with a different term added last, we found that 29% of the variance in fungal communities was attributable to host identity ($F_{52,378} = 2.89$, $R^2 = 0.24$, $P < 0.001$) and the host identity × site age interaction term ($F_{26,378} = 2.89$, $R^2 = 0.053$, $P = 0.022$), after accounting for site age. By contrast, only 6% of the variance was attributable to site age ($F_{1,378} = 5.88$, $R^2 = 0.01$, $P < 0.001$: Fig. 1b) and the site age × host identity interaction ($R^2 = 0.053$, as above) after accounting for host identity.

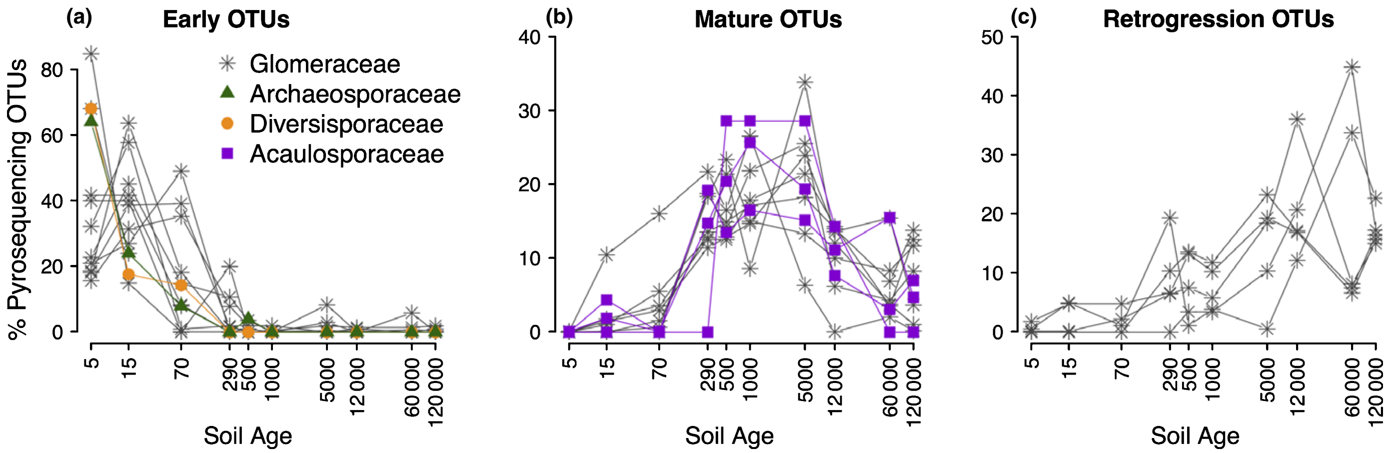

Fig. 2 Proportion of pyrosequencing operational taxonomic units (OTUs) along the Franz Josef soil chronosequence that have a significant response ($P < 0.05$) to ecosystem age: (a) linear or log-linear decreases (early OTUs), (b) quadratic (mature OTUs) and (c) linear or log-linear increases (retrogression OTUs). Model selection based on lowest Akaike Information Criterion (AIC) value.

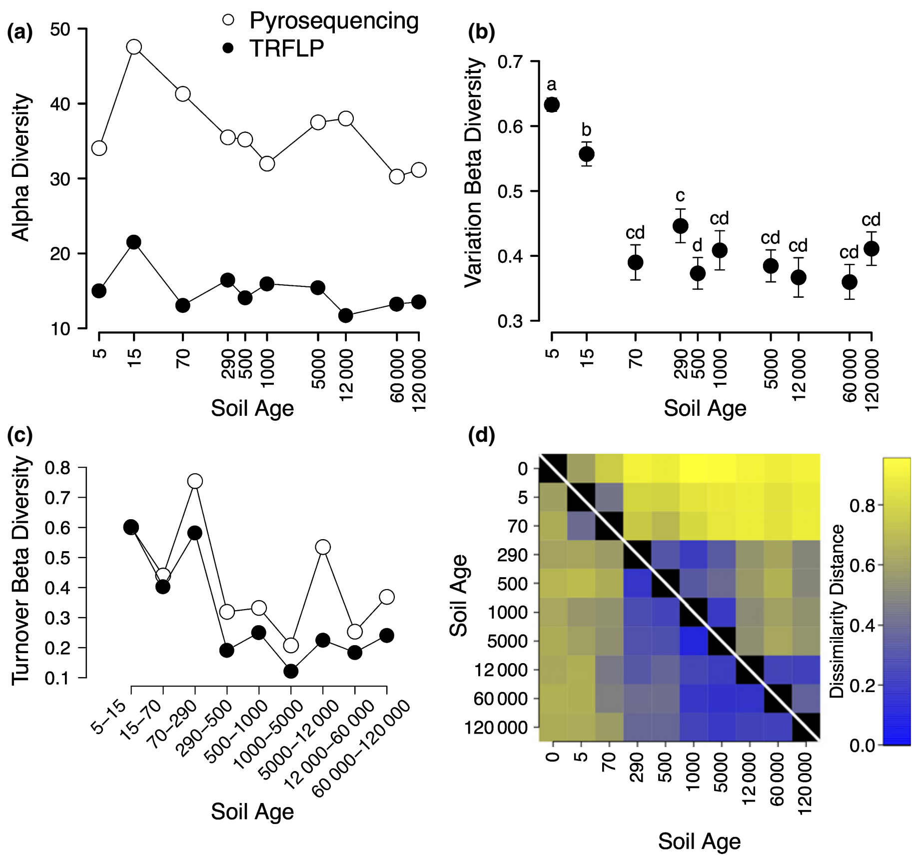

Fig. 3 (a) Rarefied alpha diversity to a sample size of 1000 and 100 for pyrosequencing and terminal restriction fragment length polymorphism (T-RFLP), respectively. (b) Variation in beta diversity (different letters indicate significant differences based on permuted P values for pairwise comparisons). (c) Turnover in beta diversity (mean ± SE). (d) Coloured matrix representing pairwise distances among all sites. Cells above and below the diagonal represent pyrosequencing and T-RFLP data, respectively. Blue–yellow colour gradient shows low–high dissimilarity distances. Bluish cells are more similar than yellow ones. All diversity measures were performed using the Bray–Curtis dissimilarity index.

To further understand the role of host plants, we tested the effect of host plant identity on arbuscular mycorrhizal fungal communities within each of the 10 site ages individually, and found a significant effect of host identity at four ages (5, 70, 60 000 and 120 000 yr; $P < 0.05$), and marginally significant effects of host identity at three additional ages (15, 5000 and 12 000 yr; $P \leq 0.1$) (Table 2). There was no significant effect of plant host identity at three sequential ages: 290, 500 and 1000 yr.

Similarly, we tested the effect of site age, independent of plant host identity, within each of the nine plant species that occurred in more than three replicates on more than one site (Table 2). Within each of these nine plant species on sites at which they occurred more than three times, only three plant species showed a significant effect of site on mycorrhizal fungal community composition. Site explained 18%, 7% and 5% of the variance in fungal communities on *Aristotelia serrata*, *Metrosideros umbellata* and

Table 2 Results from permutational multivariate analysis of variance (PerMANOVA) to test host plant identity and site age effects on arbuscular mycorrhizal fungal community composition based on terminal restriction fragment length polymorphism (T-RFLP) data

Variance partitioning by sequential sum of squares

	Factor	d.f. (factor, total)	F	P	R^2	R^2_{adj} [a]
Host identity effect after accounting for site age effect	**Site age**	**(1, 457)**	**59.86**	**0.001**	**9.6**	**9.4**
	Host identity	**(52, 457)**	**2.89**	**0.001**	**24.2**	**14.6**
	Site age × host identity	**(26, 457)**	**1.27**	**0.022**	**5.3**	**−0.7**
	Residuals	(378, 457)			60.9	
	Total				39.3	
Site age effect after accounting for host identity effect	**Host identity**	**(52, 457)**	**3.93**	**0.001**	**32.9**	**25.3**
	Site age	**(1, 457)**	**5.89**	**0.001**	**0.9**	**−0.1**
	Host identity × site age	**(26, 457)**	**1.27**	**0.019**	**5.3**	**−0.7**
	Residuals	(378, 457)			60.9	
	Total				39.3	

Host identity effect within site

Site age	d.f. (host, total)	F	P	R^2	R^2_{adj} [a]
5	**(5, 49)**	**1.44**	**0.033**	**14.0**	**2.0**
15	**(19, 46)**	**1.24**	**0.099**	**46.5**	**5.4**
70	**(8, 46)**	**1.96**	**0.014**	**29.2**	**12.0**
290	(12, 48)	1.05	0.394	26.0	−1.5
500	(10, 43)	1.44	0.114	30.4	6.5
1000	(7, 44)	1.34	0.171	20.3	2.5
5000	**(8, 42)**	**1.63**	**0.066**	**27.8**	**8.1**
12 000	**(9, 44)**	**1.54**	**0.079**	**28.4**	**7.3**
60 000	**(10, 43)**	**1.74**	**0.039**	**34.5**	**12.0**
120 000	**(10, 43)**	**2.05**	**0.007**	**38.3**	**17.1**

Site age effect within host

Host species	Sites at which the host is present 3 or more times	d.f. (site, total)	F	P	R^2	R^2_{adj} [a]
Raoulia hookeri	5/15	(1, 22)	1.67	0.102	7.4	−1.9
Aristotelia serrata	**15/70**	**(1, 17)**	**3.43**	**0.022**	**17.6**	**6.7**
Dicksonia squarrosa	290/500/1000	(1, 15)	1.51	0.194	9.8	−4.1
Metrosideros umbellata	**290/500/1000/5000/60 000**	**(1, 51)**	**3.61**	**0.009**	**6.7**	**2.9**
Weinmania racemosa	**290/500/1000/5000/12 000/60 000**	**(1, 75)**	**4.22**	**0.008**	**5.4**	**2.8**
Ripogonum scandens	500/12 000	(1, 9)	1.56	0.221	16.3	−7.6
Dacrydium cupressinum	1000/5000/12 000/60 000	(1, 19)	0.94	0.412	4.9	−6.2
Prumnopitys ferruginea	1000/5000/12 000/60 000	(1, 23)	1.24	0.306	5.3	−3.7
Metrosideros diffusa	12 000/60 000	(1, 10)	0.60	0.744	6.3	−17.2

First, using variance partitioning to disentangle host and site age effect within the overall model. Second, using individual PerMANOVAs to test the host identity effect within each site. Third, using individual PerMANOVAs to test the site age effect within the host plant species in sites in which it was found three or more times. Significant and marginal responses are shown in bold.

[a]Adjusted R^2 as an indication of effect size after adjusting for multiple factor levels [Correction added after online publication 31 December 2014: the R^2 values have been amended to correct a calculation error].

Weinmania racemosa, respectively (Table 2). The fungal communities associated with the other six plant species did not show significant effects of site age ($P > 0.1$).

In addition, variance partitioning analysis and the test of site within host were also run, treating site age as a factor instead of a continuous variable (Table S3). The model explained 46% of the total variance of fungal communities and, again, host was a strong predictor (host identity and age × host identity explaining 21% of the variance after accounting for age), and the same three plant hosts showed significant effects of site age within host.

Discussion

To our knowledge, this is the first comprehensive study of arbuscular mycorrhizal fungal communities throughout ecosystem development from progressive to retrogressive stages (Dickie *et al.*, 2013). Our study shows strong evidence of niche differentiation by fungi into different successional stages. The fungal community underwent large changes in composition over the initial 290 yr of ecosystem development, became more stable in the mature forest communities, and then underwent a second major transition between the 5000- and 12 000-yr-old sites. Despite strong shifts in soil chemistry over long-term ecosystem development (Richardson *et al.*, 2004), changes in the plant community were a strong determinant of fungal community composition, explaining 29% of the variance after accounting for site age.

Mycorrhizal fungal niche differentiation

Powell *et al.* (2009) suggested that arbuscular mycorrhizal fungal traits are phylogenetically conserved at the family level. The consistency of mycorrhizal fungal families observed here through different successional sequences suggests that these families might have functional traits selected for different stages of ecosystem development. For example, the OTUs in the families of Archaeosporaceae and Diversisporaceae showed strong preferences for early-stage communities, and the OTUs in Acaulosporaceae showed preferences for mature stages. Several other studies have also reported the presence of Diversispora species in primary succession (Wu *et al.*, 2007; Oehl *et al.*, 2011a; Sikes *et al.*, 2012), as well as *Acaulospora* species in later stages of ecosystem progression (Wu *et al.*, 2007; Oehl *et al.*, 2011a).

During retrogression, we expected to find mycorrhizal fungal families with specific adaptations to low soil P concentrations, such as the Gigasporaceae, which produce extensive external mycelium (Hart & Reader, 2002). Instead, the only fungal OTUs that increased during retrogression were in the Glomeraceae. Chagnon *et al.* (2013) suggested that the Glomeraceae were generally ruderal or pioneer species. By contrast, our findings suggest that this family is actually widespread across all ages of ecosystem development, including OTUs specialized on each ecosystem stage. Sikes *et al.* (2012) also found that *Glomus* increased in dominance through succession, albeit on a much shorter time scale. Advances in the Glomeromycota taxonomy, which is not yet fully resolved (Oehl *et al.*, 2011b; Redecker *et al.*, 2013), linked to a better knowledge of mycorrhizal fungal functional traits, will help us to understand mycorrhizal fungal distribution patterns and their function in ecosystem processes.

Mycorrhizal fungal diversity patterns across ecosystem development

Woody plant species richness increases during ecosystem progression at Franz Josef, reaching a maximum at 5000 yr (Richardson *et al.*, 2004). We had expected that arbuscular mycorrhizal fungal richness would parallel this trend and peak in mature ecosystems (Hypothesis 2). Instead, we found that alpha diversity was highest early in ecosystem development and lower in mature ecosystems. However, although mature ecosystems have a high total number of plant species (Richardson *et al.*, 2004), they also tend to be dominated by fewer, large individual plants, which dominated the sampling of roots (Fig. S3), whereas early successional communities had a more uniform size distribution of plant species, such that root sampling included a broader selection of species. Fungal OTU richness was strongly correlated with plant species richness based on the sampled roots ($F_{1,8} = 8.0$, $P = 0.0012$, $R^2_{adj} = 0.44$).

We expected a strong convergence of the mycorrhizal fungal community caused by strong host-specific mutualisms in retrogressive stages. This was not supported by the constant variation beta diversity we observed. Nevertheless, alpha diversity was lower in retrogressive stages than in early succession, and we observed high R^2 values for the host identity effect on fungal communities in retrogressive sites. Combined, these results suggest higher specificity at retrogressive stages than at previous successional stages, potentially driven by extremely low soil P concentrations (Richardson *et al.*, 2004).

Special attention has recently been given to plant–soil interactions as drivers of community assembly and diversity through ecosystem time (Fukami & Nakajima, 2013; Jangid *et al.*, 2013). Using a community simulation model, Fukami & Nakajima (2013) suggested that plant–soil interactions lengthen the time during which local plant species composition changes, and delay convergence of the plant community through time. Empirical evaluation of such models has been hindered by the lack of data on plant–microbial associations along long-term soil chronosequences. Here, we observed that early stages of succession had the most variable mycorrhizal fungal communities within and between sites of different ages, congruent with results from the primary succession of dunes (Sikes *et al.*, 2012). This high variability in early mycorrhizal fungal communities is probably a result of a lack of ecological sorting and random assembly processes.

Host identity as a driver of arbuscular mycorrhizal fungal communities

Niche differentiation by fungal species onto different soils and host plants is an important factor supporting high fungal diversity (Bruns, 1995; Dickie *et al.*, 2002; Dickie, 2007), but whether plant–fungal interactions or habitat factors are the primary drivers of fungal communities is unclear (Zobel & Öpik, 2014). This opacity is partly a result of the co-variation of plant communities and soil chemistry, which complicates the separation of factors driving fungal compositional shifts in ecosystems. In the Franz Josef chronosequence, soil chemistry and plant community covary in part with ecosystem age, but 86% of our observations came from plant species found across more than one site, enabling us to statistically disentangle the effect of plant host after accounting for site age. Both the sequential sums of squares and the analyses of host effects within sites and site effects within hosts revealed that changes in plant host were an important predictor of fungal communities. The strong host effect is surprising, given that other studies have found strong stoichiometric controls of soils on arbuscular mycorrhizal fungal communities (Lekberg *et al.*, 2007; Sikes *et al.*, 2014), and supports the Passenger/Driver hypotheses (Hart *et al.*, 2001) as opposed to the Habitat hypothesis (Zobel & Öpik, 2014).

Although arbuscular mycorrhizal fungi have been long thought to have low host specificity, there is an increasing recognition of host preference within these fungi (Vandenkoornhuyse *et al.*, 2002; Öpik *et al.*, 2009). We found evidence of a host identity effect during the first 70 yr and from 5000 to 120 000 yr; yet, from 290 to 1000 yr, there was a lack of host effect, suggesting a stronger effect of soils. Supporting this, the fungal community associated with the most commonly sampled plant species within 290- to 1000-yr-old sites (*Weinmania racemosa* and *Metrosideros umbellata*) showed a significant site effect. Zobel & Öpik (2014) suggest that the Habitat hypothesis is probably most important in mature communities in which plant and fungal dispersal is unlikely to limit species presence (i.e. stable successional communities), which is consistent with our observation. Alternatively, resource availability of both N and P are highest during this stage of ecosystem development (Richardson *et al.*, 2004), and this may result in a lower relative difference between optimal and suboptimal symbionts, thereby reducing the ability and incentive for plant discrimination for or against fungal partners (Kiers *et al.*, 2011).

Molecular methods: T-RFLP vs pyrosequencing

At present, fewer than 250 species of arbuscular mycorrhizal fungi have been described (Schüßler & Walker, 2010), representing just a subset of the estimated total diversity (Kivlin *et al.*, 2011; Öpik *et al.*, 2013). Of the 113 OTUs we described at the Franz Josef soil chronosequence, only eight match cultured mycorrhizal fungi, consistent with recent findings that natural ecosystems are composed largely of uncultured fungi (Ohsowski *et al.*, 2014). There is a strong need to conduct arbuscular mycorrhizal fungal surveys, particularly in under-studied regions, such as New Zealand, and habitats, such as temperate rainforests, to gain a better resolution of global mycorrhizal fungal diversity (Öpik *et al.*, 2013).

The use of pyrosequencing in combination with T-RFLP allowed us to maximize both the sequencing depth (37 492 sequences identified to OTUs from $n = 10$ sites) and replication ($n = 498$ root samples in T-RFLP analysis). Where we were able to directly compare the two methods (e.g. site-level responses), the results were generally congruent for alpha diversity ($R^2 = 0.33$) and community turnover ($R^2 = 0.76$). This fits a general suggestion that ecological patterns in arbuscular mycorrhizal fungal communities are robust to changing molecular approaches (Lekberg *et al.*, 2014; Powell & Sikes, 2014). Nevertheless, when comparing techniques, we found that pyrosequencing was more sensitive than T-RFLP to the fungal community shift from mature to retrogressive stages, as seen in the second peak in turnover beta diversity and in the matrix reflecting pairwise differences between sites (Fig. 3d: pyrosequencing data above the diagonal and T-RFLP data below the diagonal). This probably reflects a lower sensitivity of T-RFLP compared with pyrosequencing to within-genus shifts in fungal communities.

A recurring question in any PCR-based technique (including T-RFLP and pyrosequencing) is whether copy numbers can be treated as semi-quantitative measures of abundance (Dickie & FitzJohn, 2007; Lindahl *et al.*, 2013). We treated T-RFLP as presence/absence data at the level of individual roots, summing occurrences within a site as a measure of abundance. For pyrosequencing data, we lacked replication within site, and used DNA copy number as a measure of abundance. Despite theoretical reasons why post-PCR copy numbers should not be taken quantitatively, the congruence of T-RFLP results and pyrosequencing results is reassuring empirical evidence of validity. In any case, a strictly presence/absence approach would encounter its own difficulties as a result of tag-swapping (Carlsen *et al.*, 2012) and ecologically trivial DNA (Koele *et al.*, 2014).

Implications for mycorrhizal fungal community ecology

Most research on arbuscular mycorrhizal fungi has taken place in Petri dishes, glasshouses and agricultural fields, whereas arbuscular mycorrhizal fungi in natural ecosystems have been less well studied (Ohsowski *et al.*, 2014). This study demonstrates the value of strong ecological gradients as model systems for unravelling the complexities of the mycorrhizal symbiosis. In particular, habitat niche differentiation has previously been considered for plants and fungi as separate entities (e.g. Dickie *et al.*, 2002; McKane *et al.*, 2002; John *et al.*, 2007). Our finding that strong niche differentiation of fungi occurred largely as an effect of plant host identity, rather than soil habitat, was unexpected and provides new insights into the interaction of plants, fungi and soils (Dickie *et al.*, 2013).

Although chronosequences have great value as model systems for the understanding of mycorrhizal interactions, a better incorporation of these mycorrhizal interactions is also critical for the interpretation and understanding of the patterns of vascular plant diversity and soil nutrient availability reported from chronosequences. Mycorrhizal fungi are the fundamental link between plants and soils, playing key roles in P uptake, mineral weathering, soil aggregation and plant nutrient uptake. We suggest that a better integration of mycorrhizal functioning into chronosequence studies will help to increase our understanding of soil biogeochemistry and pedogenesis.

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Supporting Information

Additional supporting information may be found in the online version of this article.

Fig. S1 Diversity patterns of arbuscular mycorrhizal community using an operational taxonomic unit (OTU) similarity cutoff of 97%.

Fig. S2 Neighbor-Net of Franz Josef arbuscular mycorrhizal fungal community.

Fig. S3 Nonmetric multidimensional scaling (NMDS) and diversity patterns of the sampled plant community.

Table S1 Additional soil chemical properties of the different sites across Franz Josef soil chronosequence

Table S2 Number of roots sampled of each plant species at each site

Table S3 Permutational multivariate analysis of variance (PerMANOVA) to test host identity and site age effect, treating site age as a categorical variable

Notes S1 Operational taxonomic unit (OTU) clustering and recovery process.

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